Mahyar Maleki

Neuroscientist | Data Analyst | AI Engineer

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Summary

Currently a guest researcher at Freie Universität Berlin in Berlin, Germany, studying how video models and the human brain process temporal information. I hold an MSc in Artificial Intelligence from the University of Tehran and specialize in research at the intersection of AI and neuroscience. My master's thesis focused on EEG-based neural data analysis, combining cognitive neuroscience with advanced machine learning. I'm open to research and engineering positions in machine learning, computational neuroscience, and brain-inspired AI.

Education

University of Tehran — Tehran, Iran
Master of Science in Computer Engineering - Artificial Intelligence — *Sep 2021 – Sep 2024*
- Thesis title: "Comparing Representation of Video and Still Images in Brain with EEG Signals"
- Supervisor: Dr. Mohammad-Reza A. Dehaqani
- Overall GPA: 3.65/4 (17.73/20)

Imam Khomeini International University — Qazvin, Iran
Bachelor of Science in Mechanical Engineering — *Sep 2015 – Sep 2019*
- Overall GPA: 3.03/4 (15.78/20)

Research Experience

Guest Researcher — Jan 2026 – Present
Freie Universität Berlin — *Berlin, Germany*
- Investigating how temporal information shapes visual recognition, comparing video and image models with EEG and fMRI responses to naturalistic videos.
- Funded by a one-year research scholarship from Freie Universität Berlin.
- Supervised by Prof. Radoslaw M. Cichy

Research Assistant — 2022 – 2024
Convergent Technologies Research Center, University of Tehran
- Thesis: "Comparing Brain Representations of Video and Still Images using EEG Signals"
- Investigated how temporal continuity in videos affects brain representations; found that scrambling frames reduces accuracy and slows decoding, emphasizing the role of temporal information in processing dynamic stimuli.
- Supervised by Dr. Mohammad-Reza A. Dehaqani

Research Assistant — 2022 – 2023
Convergent Technologies Research Center, University of Tehran
- Contributed to the 7th phase of the "Problem-Oriented Technology Development Plan"
- Conducted research in the *"Brain-Machine Interface Development for Enhancing ASD Users' Capabilities"* project, focusing on brain data analysis and machine learning methods.
- Supervised by Dr. Mohammad-Reza A. Dehaqani

Research Interests

- **Computational Neuroscience**
- **Visual Perception**
- **Temporal Integration in Vision**
- **Video Understanding Models**
- **Explainable AI**
- **Generative AI**

Publications

- A. H. Fadaei, **M. Maleki**, M. R. A. Dehaqani. Brain-Aligned Multi-Stream Video Transformers with Sparse Self-Selection. *arXiv preprint* arXiv:2607.17625, 2026.

Teaching Experience

Teaching Assistant
Course: Trustworthy AI — *University of Tehran - Spring 2024*
- Instructor: Dr. Tavassolipour
- Designed a homework assignment on detecting and mitigating backdoor attacks, implementing methods from the "Neural Cleanse" paper.

Teaching Assistant
Course: Statistical Inference — *University of Tehran - Spring and Fall 2023*
- Instructors: Dr. Abolghasemi, Dr. Vahabie
- Mentored students in understanding key probability distributions and their practical applications using R.

Teaching Assistant
Course: Deep Neural Networks — *University of Tehran - Spring 2023*
- Instructor: Dr. Kalhor
- Created instructional videos to help students apply advanced PyTorch techniques for Convolutional Neural Networks (CNNs).

Teaching Assistant
Course: Introduction to FieldTrip — *CuttingEEG Conference - October 2023*
- Guided students on the basics of FieldTrip, a MATLAB toolbox for MEG and EEG analysis, during a hands-on workshop at the CuttingEEG conference in Tehran.

Teaching Assistant
Course: Computational Neuroscience — *Neuromatch Academy - July 2023*
- Mentored students in Machine Learning, Dynamical Systems, and Stochastic Processes, providing guidance and support for final project completion.

Selected Projects

Trustworthy AI Spring 2023
- Enhanced representation robustness using angular loss.
- Improved model interpretability with LIME and SHAP for comprehensive explanations.
- Implemented out-of-distribution detection to improve model robustness and generalization.
- Detection, Reconstruction, and Cleansing of Backdoor Triggers in CNNs.

Deep Learning and Applications Spring 2022
- Predicted saliency maps using deep neural networks.
- Developed a deep fully connected network to invert visual representations of a Convolutional Network.
- Performed medical image segmentation using U-Net architecture.
- Implemented Hubert architecture for automatic speech recognition.
- Reproduced "DeepGUM: Learning Deep Robust Regression with a Gaussian-Uniform Mixture Model".
- Classified and clustered musical instrument audio using classical machine learning methods.

Introduction to Cognitive Neuroscience Spring 2022
- Designed cognitive tasks to study spatial heterogeneity in face perception and performed participant data analysis.
- Implemented dynamical neuronal models to explore functional patterns of neurons.
- Analyzed neural data obtained via EEG, LFP, fMRI, and single-cell recordings.

Data Analysis Autumn 2021
- Analyzed COVID-19 infection data to identify trends and patterns.
- Scraped ticket prices from Alibaba and MrBilit for comparative analysis.
- Web crawling and data cleaning of laptop listings from Digikala for structured analysis.

Selected Courses

Machine Learning	4.0 (20.00/20)	**Trustworthy AI**	4.0 (17.10/20)
Advanced Robotics	4.0 (19.88/20)	**Deep Learning and Applications**	4.0 (17.00/20)
Statistical Inference	4.0 (18.20/20)	**Cognitive Neuroscience**	4.0 (16.93/20)
Data Analysis	4.0 (17.75/20)		

Honors

- Ranked 42nd in The Masters Level University Entrance Exam (Among Approximately Twelve Thousand People)
- Among the top 2% of Iran Universities Entrance Exam
- Exemption from tuition fees for master's level studies

Technical Skills

Programming: Python, Matlab, R, SQL, Git

Deep Learning Toolboxes: Pandas, NumPy, PyTorch, TensorFlow, Scikit-Learn, MMAction, Hydra/OmegaConf

Neuroscience Toolboxes: PsychoPy, Psychtoolbox, EEGLab, FieldTrip, Net2Brain, RSA

Operating Systems: Linux (particularly Ubuntu, HPC clusters), Windows

Soft Skills: Teamwork, Problem Solving, Communication, Critical Thinking, Adaptability

Languages: Persian (Native), German (Basic), English (Proficient - TOEFL Best Scores ⇒ Overall: 102, Reading: 27, Listening: 28, Speaking: 23, Writing: 24)

References

Prof. Radoslaw M. Cichy
Professor
Freie Universität Berlin / Department of Education and Psychology / Neural Dynamics of Visual Cognition
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Dr. Mohammad-Reza A. Dehaqani
Assistant Professor
University of Tehran / College of Engineering / Electrical and Computer ENG
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